

Finite Element Method for Linear Dynamics

MECH4103 — Spring 2026

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22 April 2026

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A 2nd-order PDE with two variables x_1, x_2 (with $u = u(x_1, x_2)$, $f = f(x_1, x_2)$):

$$a_{11} \frac{\partial^2 u}{\partial x_1^2} + a_{12} \frac{\partial^2 u}{\partial x_1 \partial x_2} + a_{22} \frac{\partial^2 u}{\partial x_2^2} + a_1 \frac{\partial u}{\partial x_1} + a_2 \frac{\partial u}{\partial x_2} = f$$

Classification (by analogy with conic sections):

$$a_{12}^2 - 4a_{11}a_{22} \begin{cases} < 0 & \text{elliptic} \\ = 0 & \text{parabolic} \\ > 0 & \text{hyperbolic} \end{cases}$$

Comparison of PDE Types and Stability

PDE Type	Example Equation	Stability Characteristics
Elliptic	$\nabla^2 u = f$ (Laplace / Poisson)	Always stable. FEM works well.
Parabolic	$\frac{\partial u}{\partial t} = \nabla^2 u$ (Heat Eq.)	Stable but requires proper time discretization. Explicit methods may be unstable.
Hyperbolic	$\frac{\partial^2 u}{\partial t^2} = c^2 \nabla^2 u$ (Wave Eq.)	Can be unstable without stabilization. Needs careful numerical treatment.

- Standard FEM is well-suited for **elliptic problems**, which typically yield smooth solutions.
- FEM struggles with **hyperbolic-like PDEs**. In advection-dominated problems (very small diffusion term), steep gradients may develop.
- Sharp gradients can cause **numerical instability and spurious oscillations** if not addressed with stabilization techniques.

Advection-dominated limit ($a_{11} \approx 0$, $a_{22} \approx 0$)

$$b_1 \frac{\partial u}{\partial x} + b_2 \frac{\partial u}{\partial y} = f$$

The equation behaves like a *hyperbolic PDE*. Stabilization methods such as **Galerkin Least Squares (GLS)** are required to suppress oscillations.

* Stabilization methods modify the weak form to suppress oscillations without sacrificing accuracy.

Static FEM assumed:

- Small deformations, linear material behaviour
- **No inertia or damping**

Linear Dynamics extends this by incorporating:

- Inertia terms (and optionally damping)
- Time derivatives in the governing equations

This enables two types of analyses:

- **Steady-state motion** — harmonic responses: free vibration (natural frequencies) and forced vibration (frequency response)
- **Transient motion** — time-varying responses via direct time integration or mode superposition

Continuous fields

$\mathbf{u}$, $\boldsymbol{\sigma}$, $\boldsymbol{\varepsilon}$ — vector/matrix fields

$\mathbf{D}$ — constitutive matrix

∇_S — symmetric gradient operator

Discretized quantities

$\mathbf{N}$ — shape functions

$\mathbf{M}$, $\mathbf{K}$, $\mathbf{F}$ — system matrices/vectors

$\dot{\mathbf{a}}$ — velocity, $\ddot{\mathbf{a}}$ — acceleration

1 Equilibrium (with inertia):

$$\nabla_S^T \boldsymbol{\sigma} + \mathbf{b} = \rho \ddot{\mathbf{u}} \quad \text{on } \Omega$$

ρ — mass density, $\ddot{\mathbf{u}} = [\ddot{u}_x, \ddot{u}_y]^T$

2 Constitutive equation:

$$\boldsymbol{\sigma} = \mathbf{D} \nabla_S \mathbf{u}$$

3 Boundary conditions:

- Traction: $\mathbf{t} = \boldsymbol{\tau} \mathbf{n}$ on Γ_t
- Displacement: $\mathbf{u} = \bar{\mathbf{u}}$ on Γ_u

4 Initial conditions:

$$\mathbf{u}(x, 0) = \mathbf{u}_0(x), \quad \dot{\mathbf{u}}(x, 0) = \mathbf{v}_0(x)$$

Multiply equilibrium by test functions $\mathbf{w} \in U_0$ and integrate over Ω ; apply Green's theorem:

$$\underbrace{\int_{\Omega} (\nabla_S \mathbf{w})^T \mathbf{D} \nabla_S \mathbf{u} d\Omega}_{\text{stiffness term}} + \underbrace{\int_{\Omega} \mathbf{w}^T \rho \ddot{\mathbf{u}} d\Omega}_{\text{inertia term}} = \underbrace{\int_{\Gamma_t} \mathbf{w}^T \bar{\mathbf{t}} d\Gamma + \int_{\Omega} \mathbf{w}^T \mathbf{b} d\Omega}_{\text{external work}}$$

for all $\mathbf{w} \in U_0 = \{\mathbf{w} \mid \mathbf{w} \in H^1, \mathbf{w} = 0 \text{ on } \Gamma_u\}$.

Approximate: $\mathbf{u} \approx \mathbf{N}\mathbf{a}$, $\mathbf{w} = \mathbf{N}\mathbf{c}$, $\nabla_S \mathbf{u} = \mathbf{B}\mathbf{a}$

Substituting into the weak form yields the **semi-discrete system**:

$$\mathbf{M} \ddot{\mathbf{a}} + \mathbf{K} \mathbf{a} = \mathbf{F}(t)$$

System matrices

$$\mathbf{M} = \int_{\Omega} \mathbf{N}^T \rho \mathbf{N} d\Omega, \quad \mathbf{K} = \int_{\Omega} \mathbf{B}^T \mathbf{D} \mathbf{B} d\Omega, \quad \mathbf{F}(t) = \int_{\Gamma_t} \mathbf{N}^T \bar{\mathbf{t}} d\Gamma + \int_{\Omega} \mathbf{N}^T \mathbf{b} d\Omega$$

For $\mathbf{F} = 0$, assume harmonic solution $\mathbf{a}(t) = A\mathbf{\Phi} e^{i\omega t}$:

$$(\mathbf{K} - \omega^2 \mathbf{M})\mathbf{a} = \mathbf{0}$$

Non-trivial solutions require:

$$\det(\mathbf{K} - \omega^2 \mathbf{M}) = 0 \quad \implies \quad \text{eigenvalues } \omega_i^2, \quad \text{eigenvectors } \mathbf{\Phi}_i$$

- ω_i — natural frequencies
- $\mathbf{\Phi}_i$ — mode shapes (spatial deformation patterns)
- Amplitude A determined by initial conditions; does not affect the eigenvalue problem

Direct computation of $\det(\mathbf{K} - \omega^2 \mathbf{M}) = 0$ is infeasible for large systems. Standard approach:

- 1 Rewrite as standard eigenvalue problem:

$$\mathcal{K} \mathbf{a} = \omega^2 \mathbf{a}, \quad \mathcal{K} = \mathbf{M}^{-1} \mathbf{K}$$

- 2 **Preferred: inverse formulation** (focus on low frequencies):

$$\mathbf{K}^{-1} \mathbf{M} \mathbf{a} = \frac{1}{\omega^2} \mathbf{a} \quad \Rightarrow \quad \omega^2 = 1/\lambda, \quad \lambda = \text{eig}(\mathbf{K}^{-1} \mathbf{M})$$

Why the inverse formulation?

- Smallest $\omega^2 \leftrightarrow$ largest $1/\omega^2$ — easier to find numerically
- Better numerical stability; exploits sparsity of $\mathbf{K}$, $\mathbf{M}$

For harmonic loading $\mathbf{F}(t) = \mathbf{F}_0 e^{i\omega t}$, assume $\mathbf{a}(t) = \mathbf{A} e^{i\omega t}$:

$$(\mathbf{K} - \omega^2 \mathbf{M}) \mathbf{A} = \mathbf{F}_0$$

Solve for $\mathbf{A}$ — the **frequency response amplitude**.

- Resonance occurs when $\omega \approx \omega_i$ (natural frequency)
- Near resonance, $\mathbf{K} - \omega^2 \mathbf{M}$ becomes (near-)singular
- Damping regularizes the response (see Section on Damped Systems)

Discretize time $t_n = n \Delta t$ and march forward in time.

Explicit methods

- Compute $\mathbf{a}_{n+1}$ from known values $\mathbf{a}_n$, $\mathbf{a}_{n-1}$
- Efficient per step
- **Conditionally stable** (Δt must be small)

Implicit methods

- Solve a system involving $\mathbf{a}_{n+1}$
- More effort per step
- Often **unconditionally stable**

Central Difference Method (Explicit)

Goal: given knowns at t_n and t_{n-1} , find $\mathbf{a}_{n+1}$.

Knowns at step n

- $\mathbf{a}_{n-1}$, $\mathbf{a}_n$ — displacements
- $\dot{\mathbf{a}}_n$, $\ddot{\mathbf{a}}_n$ — vel. & accel.
- $\mathbf{F}_n$ — external force

Unknown at step $n + 1$

- $\mathbf{a}_{n+1}$ — displacement only
- (velocity and acceleration are *not* needed at $n + 1$ to advance the solution)

Step 1 — Equation of motion satisfied at t_n :

$$\mathbf{M} \ddot{\mathbf{a}}_n + \mathbf{K} \mathbf{a}_n = \mathbf{F}_n$$

Step 2 — Replace $\ddot{\mathbf{a}}_n$ with finite differences:

$$\dot{\mathbf{a}}_n \approx \frac{\mathbf{a}_{n+1} - \mathbf{a}_{n-1}}{2\Delta t}, \quad \ddot{\mathbf{a}}_n \approx \frac{\mathbf{a}_{n+1} - 2\mathbf{a}_n + \mathbf{a}_{n-1}}{\Delta t^2}$$

Step 3 — Solve explicitly for the only unknown $\mathbf{a}_{n+1}$:

$$\mathbf{a}_{n+1} = \mathbf{M}^{-1} [\Delta t^2 (\mathbf{F}_n - \mathbf{K} \mathbf{a}_n) + 2\mathbf{M} \mathbf{a}_n - \mathbf{M} \mathbf{a}_{n-1}]$$

Initialisation (start-up values)

Goal: given knowns at t_n , find $\mathbf{a}_{n+1}$ by satisfying the equation of motion at t_{n+1} .

Knowns at step n

- $\mathbf{a}_n$ — displacement
- $\dot{\mathbf{a}}_n$ — velocity
- $\ddot{\mathbf{a}}_n$ — acceleration
- $\mathbf{F}_{n+1}$ — external force at t_{n+1}

Unknowns at step $n + 1$

- $\mathbf{a}_{n+1}$ — displacement
- $\dot{\mathbf{a}}_{n+1}$ — velocity
- $\ddot{\mathbf{a}}_{n+1}$ — acceleration

Step 1 — Equation of motion at t_{n+1} (must be satisfied):

$$\mathbf{M} \ddot{\mathbf{a}}_{n+1} + \mathbf{K} \mathbf{a}_{n+1} = \mathbf{F}_{n+1}$$

3 unknowns, 1 equation $\Rightarrow$ two more relations needed.

Step 2 — Predictor equations (interpolate acceleration over Δt):

$$\dot{\mathbf{a}}_{n+1} = \dot{\mathbf{a}}_n + (1 - \gamma)\Delta t \ddot{\mathbf{a}}_n + \gamma\Delta t \ddot{\mathbf{a}}_{n+1}$$

$$\mathbf{a}_{n+1} = \mathbf{a}_n + \Delta t \dot{\mathbf{a}}_n + \left(\frac{1}{2} - \beta\right)\Delta t^2 \ddot{\mathbf{a}}_n + \beta\Delta t^2 \ddot{\mathbf{a}}_{n+1}$$

β and γ control how much of $\ddot{\mathbf{a}}_{n+1}$ is used in the update.

Step 3 — Express $\ddot{\mathbf{a}}_{n+1}$ in terms of $\mathbf{a}_{n+1}$ using the *displacement* predictor only:

Rearrange the displacement predictor:

$$\ddot{\mathbf{a}}_{n+1} = \frac{1}{\beta \Delta t^2} \left[\mathbf{a}_{n+1} - \mathbf{a}_n - \Delta t \dot{\mathbf{a}}_n - \left(\frac{1}{2} - \beta \right) \Delta t^2 \ddot{\mathbf{a}}_n \right] = \frac{\mathbf{a}_{n+1} - \tilde{\mathbf{a}}}{\beta \Delta t^2}$$

where all known quantities are collected into:

$$\tilde{\mathbf{a}} = \mathbf{a}_n + \Delta t \dot{\mathbf{a}}_n + \left(\frac{1}{2} - \beta \right) \Delta t^2 \ddot{\mathbf{a}}_n$$

Step 4 — Substitute into the equation of motion and collect $\mathbf{a}_{n+1}$:

$$\mathbf{M} \frac{\mathbf{a}_{n+1} - \tilde{\mathbf{a}}}{\beta \Delta t^2} + \mathbf{K} \mathbf{a}_{n+1} = \mathbf{F}_{n+1}$$

$$\underbrace{\left(\mathbf{M} + \beta \Delta t^2 \mathbf{K} \right)}_{\hat{\mathbf{K}}} \mathbf{a}_{n+1} = \underbrace{\mathbf{F}_{n+1} + \frac{\mathbf{M} \tilde{\mathbf{a}}}{\beta \Delta t^2}}_{\hat{\mathbf{F}}_{n+1}}$$

Solve this linear system once per time step to obtain $\mathbf{a}_{n+1}$, then recover $\ddot{\mathbf{a}}_{n+1}$ and $\dot{\mathbf{a}}_{n+1}$ from the predictor equations.

Express solution as a linear combination of mode shapes:

$$\mathbf{a}(t) = \sum_{i=1}^n \mathbf{\Phi}_i q_i(t)$$

where $\mathbf{\Phi}_i$ solve $(\mathbf{K} - \omega_i^2 \mathbf{M})\mathbf{\Phi}_i = \mathbf{0}$ and $q_i(t)$ are modal coordinates.

Key advantage: orthogonality of modes decouples the system into independent scalar ODEs:

$$\ddot{q}_j + \omega_j^2 q_j = \frac{\mathbf{\Phi}_j^T \mathbf{F}(t)}{\mathbf{\Phi}_j^T \mathbf{M} \mathbf{\Phi}_j}, \quad j = 1, \dots, n$$

Table: Recommended method based on problem characteristics.

Situation	Recommended Method
Long simulation, low-frequency load, linear system	Mode superposition
Short simulation or impulsive load	Direct integration
Nonlinear material or geometry	Direct integration only
You already need natural frequencies for design	Mode superposition (reuse the eigen-solve)

Including damping via matrix $\mathbf{C}$:

$$\mathbf{M} \ddot{\mathbf{a}} + \mathbf{C} \dot{\mathbf{a}} + \mathbf{K} \mathbf{a} = \mathbf{F}(t)$$

Rayleigh damping (proportional damping):

$$\mathbf{C} = \alpha \mathbf{M} + \beta \mathbf{K}$$

- α — mass-proportional coefficient (attenuates low frequencies)
- β — stiffness-proportional coefficient (attenuates high frequencies)
- Preserves the banded/sparse structure of the system matrices

Damping affects all analyses (free vibration, frequency response, transient) but is omitted in the derivations above for clarity.

- PDE type (elliptic / parabolic / hyperbolic) governs FEM stability; advection-dominated problems require stabilization (e.g. GLS).
- Linear dynamics adds inertia $\mathbf{M}\ddot{\mathbf{a}}$ to the static system $\mathbf{K}\mathbf{a} = \mathbf{F}$.
- **Free vibration** $\Rightarrow$ generalized eigenvalue problem; inverse formulation preferred numerically.
- **Forced vibration** $\Rightarrow$ frequency response $(\mathbf{K} - \omega^2\mathbf{M})\mathbf{A} = \mathbf{F}_0$.
- **Transient**: explicit Central Difference (conditionally stable) vs. implicit Newmark (unconditionally stable for proper parameters).
- **Mode superposition**: efficient for low-frequency-dominated linear problems; limited for nonlinear or impulsive loads.
- Rayleigh damping $\mathbf{C} = \alpha\mathbf{M} + \beta\mathbf{K}$ is a practical proportional-damping model.